S3.2 Functional groups: Classification of organic compounds

Proton nuclear magnetic resonance spectroscopy (^{1}H NMR) (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to:

- (S3.2.10) Interpret ^{1}H NMR spectra to deduce the structures of organic molecules from the number of signals, the chemical shifts, and the relative areas under signals (integration traces).
- (S3.2.11) Interpret ^{1}H NMR spectra from splitting patterns showing singlets, doublets, triplets and quartets to deduce greater structural detail.

Magnetic resonance imaging (MRI) is a technique that uses a magnetic field to produce a detailed view of the inside of the human body. But how does it work? The human body is made up of approximately 60% water, which is found in all the major organs of the human body as shown in **Figure 1**.

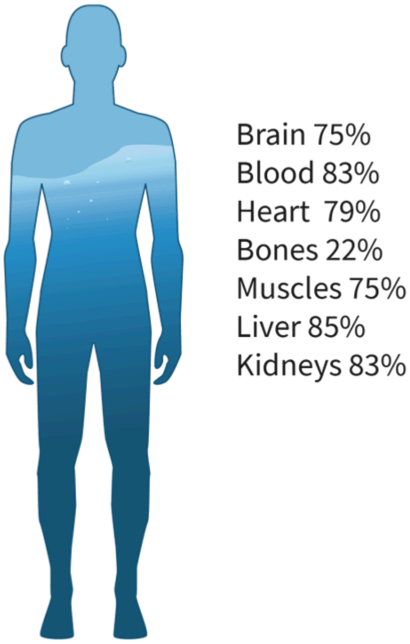


Figure 1. Many of the organs in the human body have a high percentage of water: this property is used to produce an MRI scan.

More information for figure 1

The image shows a silhouette of a human figure filled with a blue gradient, representing the body's water content. To the right of the silhouette, there is a list of organs along with their respective water percentages: Brain 75%, Blood 83%, Heart 79%, Bones 22%, Muscles 75%, Liver 85%, Kidneys 83%. The silhouette visually aligns with the mentioned percentages, indicating how much of each organ is composed of water. This representation helps explain the mechanism of MRI scans, which utilize the body's water content for imaging.

[Generated by AI]

Water consists of two hydrogen atoms and one oxygen atom, and each hydrogen atom has one proton in its nucleus. When placed in a strong magnetic field, these protons behave like tiny bar magnets and can align with or against the magnetic field. Radio waves are then used to flip the protons that absorb the radiation. This radiation is then re-emitted and detected by sensors in the MRI scanner, which produces an MRI scan that can be used to diagnose certain medical conditions (**Figure 2**).

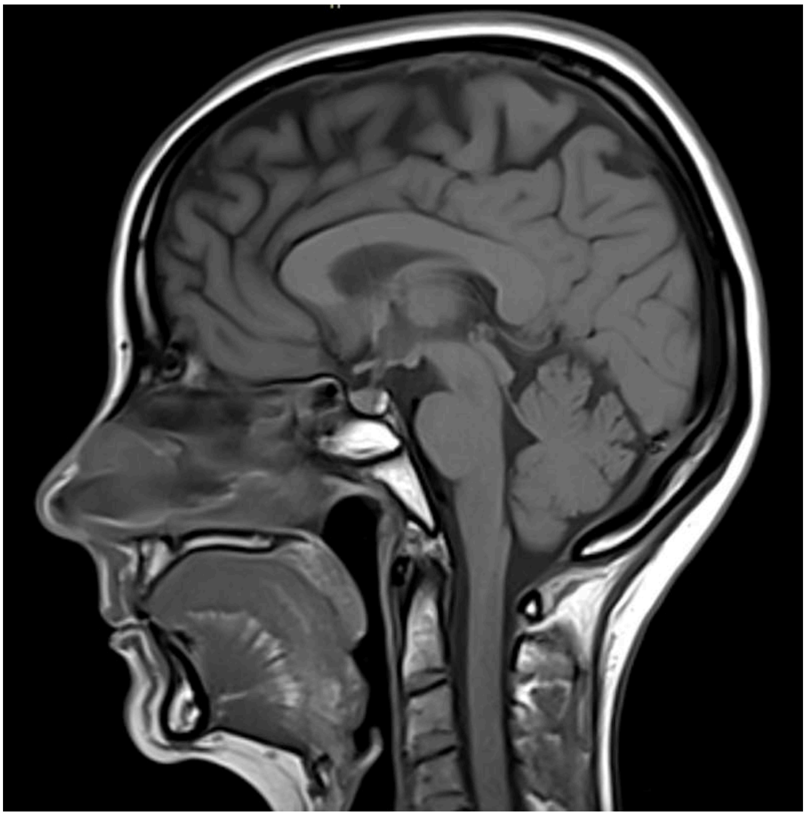


Figure 2. An MRI scan of the human brain.

Credit: Paul Biris, Getty Images

Video 1. How an MRI scanner works.

The basis of the technique of MRI is proton nuclear magnetic resonance, which is usually abbreviated to ^{1}H NMR. This is a valuable analytic technique that is used to determine the structure of unknown organic compounds. Like all subatomic particles, protons spin on their axis, which creates a magnetic field. In particles with odd numbers of nucleons, such as ^{1}H, the spins do not cancel out. When placed in a strong magnetic field, these protons behave like tiny bar magnets and can align with or against the external magnetic field (**Figure 3**).

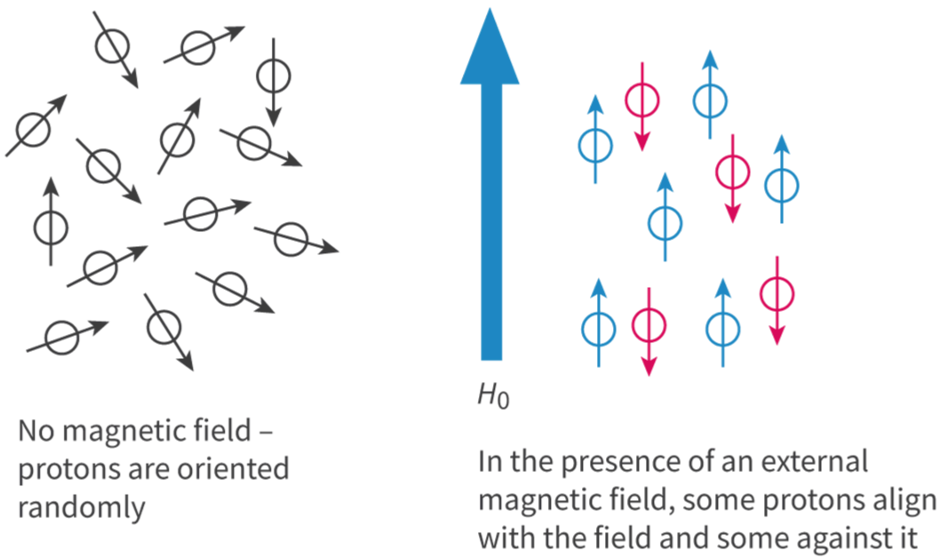


Figure 3. The effect of an external magnetic field on protons. Some align with the magnetic field and some against it.

More information for figure 3

The diagram illustrates the effect of a magnetic field on proton alignment. On the left side, protons are shown as circles with arrows pointing in random directions, labeled as 'No magnetic field — protons are oriented randomly.' On the right side, a large vertical arrow labeled 'H0' indicates the presence of an external magnetic field. Protons, depicted as circles with arrows, are shown aligned in parallel and anti-parallel orientations relative to the magnetic field. Text below reads, 'In the presence of an external magnetic field, some protons align with the field and some against it.' The two different orientations of protons suggest the effect of the magnetic field on their alignment.

[Generated by AI]

The two orientations are known as spin states. The protons aligned against the magnetic field have a slightly higher energy than those aligned with the magnetic field. However, the difference in energy between the two spin states is relatively small and can be provided by radio waves (**Figure 4**).

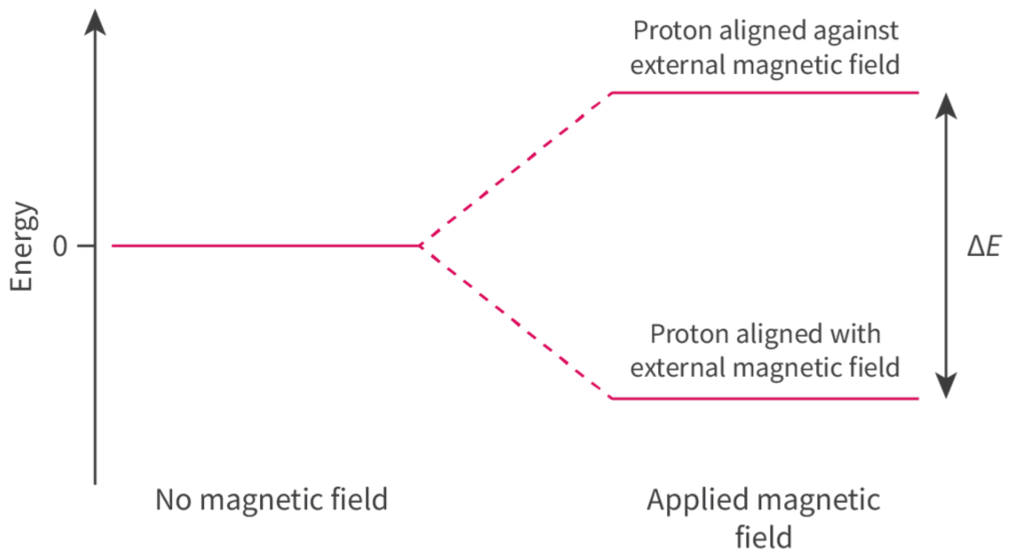


Figure 4. The relative energies of the two spin states: protons aligned with the external magnetic field have a lower energy than those aligned against it.

More information for figure 4

The diagram illustrates the energy levels of protons in relation to a magnetic field, showing two distinct states. On the left side, labeled 'No magnetic field,' there is a single horizontal line at energy level 0, indicating no differentiation in energy. On the right side, labeled 'Applied magnetic field,' two dashed lines diverge upwards from the energy level 0. One line, labeled 'Proton aligned against external magnetic field,' ascends to a higher energy state. The other line, labeled 'Proton aligned with external magnetic field,' descends to a lower energy state. The vertical double-headed arrow between these two lines is labeled 'ΔE,' indicating the energy difference between the two states.

[Generated by AI]

If the protons aligned with the magnetic field are exposed to a radio frequency signal of a specific frequency, known as the resonance frequency, some of these protons flip from the lower energy state to the higher energy state. When the radio frequency signal is switched off, these protons will return to the lower energy state and re-emit the same frequency of radiation that was absorbed. This radiation is then detected by the ^{1}H NMR spectrometer producing a ^{1}H NMR spectrum.

^{1}H NMR is used to determine the number of different chemical environments that contain protons in a molecule, which gives clues to its structure. Determining the number of different chemical environments that contain protons in an organic compound can be a challenge. However, there is a simple method, which is to look for a line of symmetry in the molecule. Using ethanal as an example (**Figure 5**), we can see that the molecule is asymmetric so the protons in the CH_3 group and the one proton in the CHO group are in different chemical environments. Protons in the same chemical environment are known as equivalent protons and those in different chemical environments are known as non-equivalent protons.

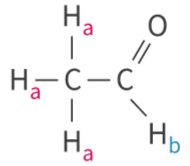


Figure 5. Ethanal has two different chemical environments that contain protons labelled H_a and H_b.

More information for figure 5

The image is a chemical diagram of ethanal (acetaldehyde) that illustrates two different chemical environments containing protons. In this molecule, a carbon atom is double-bonded to an oxygen atom, forming a carbonyl group. A hydrogen atom (labeled Hb) is bonded to the carbon atom that is also connected to the oxygen. The other carbon atom is bonded to three hydrogen atoms, each labeled Ha. The diagram notes that Ha protons are equivalent (since they are in the same chemical environment within the CH3 group), whereas the Hb proton is in a different environment, making them non-equivalent. This is visually represented by the different labels used in combination with bonds indicating their positions on the molecular structure.

[Generated by AI]

The next example we will look at is ethane (**Figure 6**). Ethane has a line of symmetry, which means that the six protons in the two CH_3 groups are equivalent protons (in the same chemical environment).

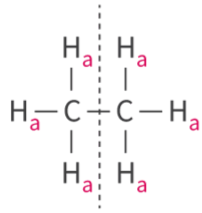


Figure 6. Ethane is symmetrical meaning that all the protons are in the same chemical environment.

More information for figure 6

This diagram illustrates the chemical structure of ethane. It displays two carbon atoms (C) in the center connected by a single bond. Each carbon atom is bonded to three hydrogen atoms (H). The diagram highlights a line of symmetry through the middle, indicating the molecule's symmetrical nature. All the hydrogen atoms are labeled as 'H_a,' showing that they are in equivalent chemical environments. This represents the symmetrical nature of ethane, where all protons are identical.

[Generated by AI]

Next, we will look at the compound butanoic acid. There is no line of symmetry in the molecule so each set of protons is in a different chemical environment (**Figure 7**). Note that the proton in the OH group is also in a separate chemical environment from the other protons.

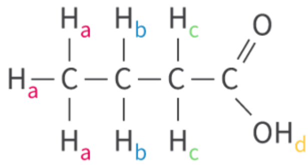


Figure 7. Butanoic acid has four different chemical environments that contain protons.

More information for figure 7

The image displays the chemical structure diagram of butanoic acid. It is depicted as a chain of four carbon atoms, each with attached hydrogen atoms, arranged linearly. The first carbon atom is connected to three hydrogen atoms, labeled as 'H_a'. The second carbon atom is connected to two hydrogen atoms, also labeled as 'H_a'. The third carbon atom is connected to two hydrogen atoms, labeled as 'H_b'. The fourth carbon atom is double-bonded to an oxygen atom (O) and single-bonded to a hydroxyl group (OH). The proton in the hydroxyl group is labeled as 'H_d'. Each different label represents a unique chemical environment for the protons, illustrating that there are multiple environments within the molecule of butanoic acid.

[Generated by AI]

Next, we will look at the butan-2-one (**Figure 8**). Try to determine the number of different chemical environments that contain protons. Check your answer when you are ready.

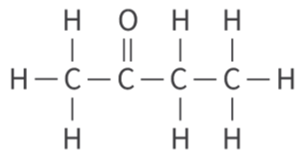


Figure 8. How many different chemical environments that contain protons are there in butan-2-one?

More information for figure 8

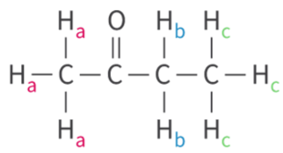
The image shows the structural formula of butan-2-one, an organic compound. It consists of a four-carbon chain with the molecular formula C4H8O. In the structure:

- The leftmost carbon atom is connected to three hydrogen atoms (CH3).
- The second carbon is connected to two hydrogen atoms (CH2) and double-bonded to an oxygen atom.
- The third carbon is connected to one hydrogen atom and a methyl group (CH2).
- The rightmost carbon atom is also connected to three hydrogen atoms (CH3).

The double bond with oxygen is located between the second and third carbon atoms, indicating the ketone group. Such a structure suggests different chemical environments related to the hydrogen atoms attached to varying parts of the molecule.

[Generated by AI]

Butan-2-one is asymmetric so each set of protons is non-equivalent. It has three different chemical environments that contain protons.



Worked example 1

Deduce the number of different chemical environments that contain protons in propane.

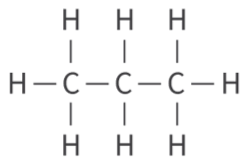


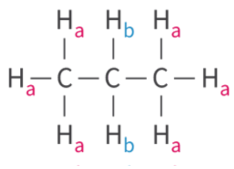
Figure 9. Propane.

More information for figure 9

The image is a diagram showing the molecular structure of propane. The structure consists of three carbon atoms connected in a straight chain. The first carbon atom on the left is bonded to three hydrogen atoms. The second carbon atom in the middle is bonded to two hydrogen atoms, one above and one below the carbon chain. The third carbon atom on the right is also bonded to three hydrogen atoms. All bonds between the carbon and hydrogen atoms are represented by straight lines, and the overall shape of the structure is linear. This diagram visually represents the arrangement of atoms in the propane molecule, highlighting both the carbon-carbon and carbon-hydrogen bonds.

[Generated by AI]

Propane has a line of symmetry so the six protons in the two CH_3 groups are equivalent and in the same chemical environment. The two protons in the CH_2 groups are also equivalent. This gives a total of two different chemical environments that contain protons.



Worked example 2

Deduce the number of different chemical environments that contain protons in ethoxyethane.

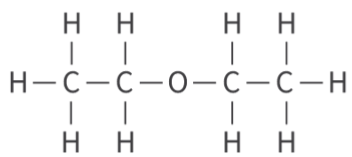


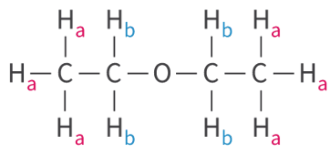
Figure 10. Ethoxyethane.

More information for figure 10

The image depicts the molecular structure of ethoxyethane, also known as diethyl ether. It is shown as a chemical diagram with labeled atoms and bonds. The structure consists of two carbon (C) atoms each bonded to three hydrogen (H) atoms on either end. These carbon atoms are connected to an oxygen (O) atom in the center, forming a symmetrical molecular structure. The connections represent single covalent bonds between the atoms. This structural arrangement indicates the presence of different types of chemical environments for protons within the molecule, based on the positioning of the hydrogen atoms around the carbon and the presence of the ether oxygen linkage.

[Generated by AI]

Ethoxyethane has two different chemical environments that contain protons. The protons in the two CH_3 groups are equivalent as are the protons in the two CH_2 groups.



Interpreting ¹H NMR spectra

A ^{1}H NMR spectrum can be used to determine the following:

1. The number of different chemical environments that contain protons, which is shown by the number of signals on the spectrum.
2. The type of protons in each different chemical environment, which is shown by the chemical shift of the signal.
3. The ratio of protons in each different chemical environment, which is shown by the integration trace.
4. The number of adjacent hydrogens is shown by splitting patterns that will be covered later in this section.

The low resolution ^{1}H NMR spectrum of ethanal is shown in **Figure 11.** The *x*-axis of a ^{1}H NMR spectrum is labelled as chemical shift, symbol δ, in ppm (parts per million), which is measured relative to tetramethylsilane, or TMS. **Table 1** showing the chemical shifts can be found in section 21 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/%C2%B9h-nmr-data-id-45123/) of the DP chemistry data booklet, and is reproduced below.

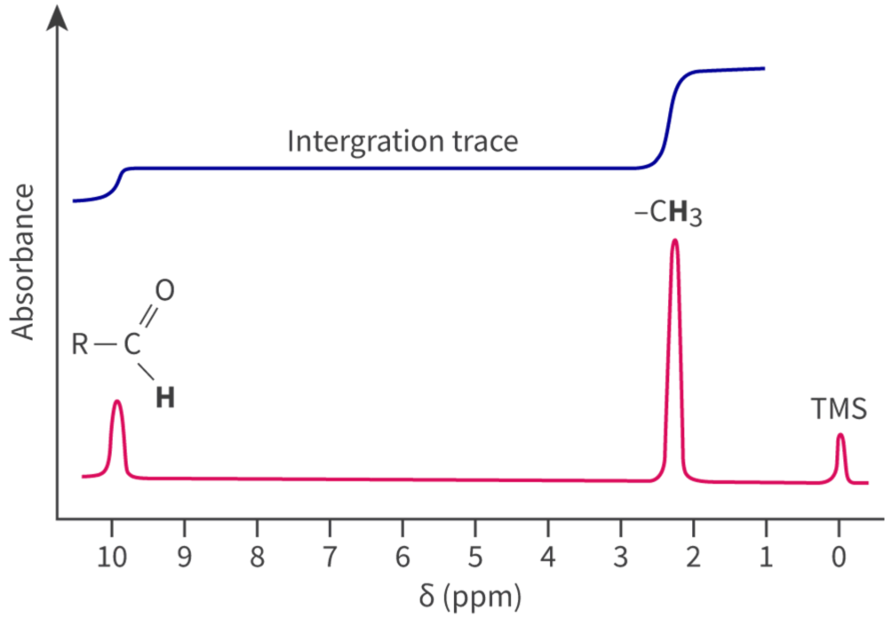


Figure 11. The ^{1}H NMR of ethanal together with an integration trace.

More information for figure 11

The image depicts a low-resolution 1H NMR spectrum of ethanal. The x-axis is labeled with the chemical shift symbol δ in ppm (parts per million) and ranges from 0 to 10 ppm, increasing from right to left. The y-axis, representing absorbance, is not explicitly labeled with values. Three key peaks are visible:

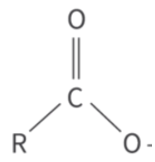
1. A low peak around 9.7 ppm marked with the structure 'R-C=O'.
2. A high peak at approximately 2.2 ppm labeled '-CH3'.
3. A smaller peak near 0 ppm labeled 'TMS'.

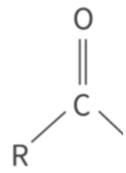
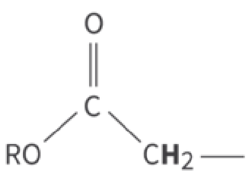
Additionally, there's a blue integration trace above the peaks. It starts at the baseline, rises between the 9.7 ppm peak and 2.2 ppm peak, levels off, and then rises again before the 0 ppm peak, indicating the relative number of protons contributing to each signal.

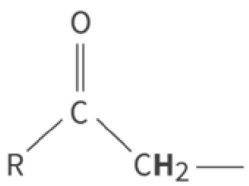
[Generated by AI]

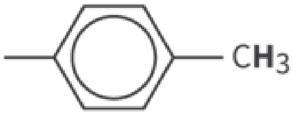
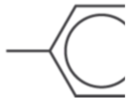
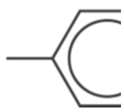
Table 1. Protons and their chemical shifts

Type of proton	Chemical shift (ppm)	Type of
$—CH_3$	0.9–1.0	R—O—

Type of proton	Chemical shift (ppm)	Type of
—CH_2—R	1.3–1.4	 More i The image chemical st diagram re an organic ester. The c consists of following component central carl (C) double- an oxygen located at t the left, the atom is sing bonded to represente which is a g symbol for aryl group. right, the c atom is sing bonded to atom (O), w further bon CH2 group overall stru depicts an functional g characteriz R-C-O-CH sequence. [Generated

Type of proton	Chemical shift (ppm)	Type of
$—R_2CH$	1.5	 More i The image structural d a carboxyli functional g features a c carbon ato is double-b an oxygen a on the top, the carbony The same c atom is sing bonded to a group (-OH right. Addit carbon ato bonded to a R group on bottom left, representin of the mole which this f group is att This depicti standard w illustrating acids in che [Generated
	2.0–2.5	R—C

Type of proton	Chemical shift (ppm)	Type of
More information The image is a chemical structure diagram with a central carbon (C) atom double bonded to an oxygen (O) atom above it. The carbon atom is also single bonded to a variable group denoted as R on the left, and to a CH2 group on the right. The R group can vary in chemical structures, while CH2 indicates a methylene group, which is a common component in organic chemistry. This type of diagram is typically used to represent organic compounds. [Generated by AI]		
 More information The image is a chemical structure diagram with a carbon atom (C) centrally connected to three groups. There is a double bond to an oxygen atom (O) at the top, a single bond to a general R group or hydrocarbon chain on the left, and a single bond to a methylene group (CH2) on the right, indicating a generic structure of a ketone or similar compound. [Generated by AI]	2.2–2.7	—HC

Type of proton	Chemical shift (ppm)	Type of
 More information The image is a chemical structure diagram representing the molecule of toluene. It features a benzene ring, which is a hexagonal shape with a circle inside, illustrating aromaticity. This ring is connected to a methyl group denoted by "CH3" on the right side. The main components are the hexagonal benzene ring at the center and the methyl group extending horizontally to the right. The diagram highlights the organic structure, with lines indicating bonds, but no colors or additional elements are present. [Generated by AI]	2.5−3.5	 More i The image chemical st typically kn phenol. It c a hexagona ring shown sided polyg alternating bonds repr a circle insi hexagon. A group, or h group, is at one of the c atoms in th ring, extenc outward to The structu highlights t component phenol and commonly represent it molecular configuratio diagrams. [Generated
—C≡C—H	1.8−3.1	 More i

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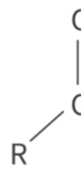
Type of proton	Chemical shift (ppm)	Type of
		The image hexagonal structure re benzene. Tl hexagon is of six carbc connected alternating double bon forming a r Attached tc a single hyc atom, repre the letter 'H connected the carbon The hexagc centered, a hydrogen a the right sic structure. [Generated
$—CH_2—Hal$	3.5–4.4	 More ir The image structural d an organic At the cent carbon ator 'C'. It forms bond with a atom label positioned To the bott carbon is si bonded to a group repre

Type of proton	Chemical shift (ppm)	Type of
		'R'. To the b right, there bond to a h atom labele Lines repre bonds betw atoms, dem the connec molecule. [Generated

In the ^{1}H NMR spectrum of ethanal, we see two signals that correspond to the two different chemical environments that contain protons: one at approximately 2 ppm and another at approximately 10 ppm (we ignore the signal at 0 ppm as it is caused by the reference standard TMS). Using **Table 1**, we can determine that the signal at 10 ppm is produced by the proton bonded to the carbonyl group in an aldehyde (–C**H**O). The signal at 2 ppm is produced by the three protons in a methyl group, –C$\mathbf{H_3}$.

Study skills

According to the table, this signal for the protons in the CH_3 group should appear at 0.9—1.0 ppm. However, the presence of the electronegative oxygen atom in close proximity causes the signal to move downfield: this is known as deshielding.

In n ^{1}H NMR spectrum, the area under each signal is significant as it can be used to determine the ratio of protons in each different chemical environment. The area under each signal is represented by an integration trace. An integration trace goes up in 'steps' that are proportional to the number of protons in each different chemical environment. **Figure 12** shows the isolated integration trace of ethanal. Note that the height of the steps in the integration trace is the ratio of the protons in each different chemical environment, which is 1:3.

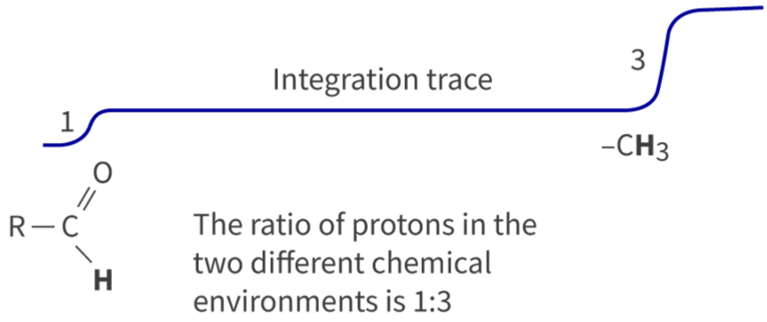


Figure 12. The integration trace of ethanal shows the ratio of the protons in the different chemical environments.

More information for figure 12

Worked example 3

The ^{1}H NMR spectrum of a compound is shown below. Deduce the following:

1. The number of different chemical environments that contain protons.
2. The ratio of protons in each chemical environment.
3. The protons responsible for the signals at chemical shift 1.0 and 5.2

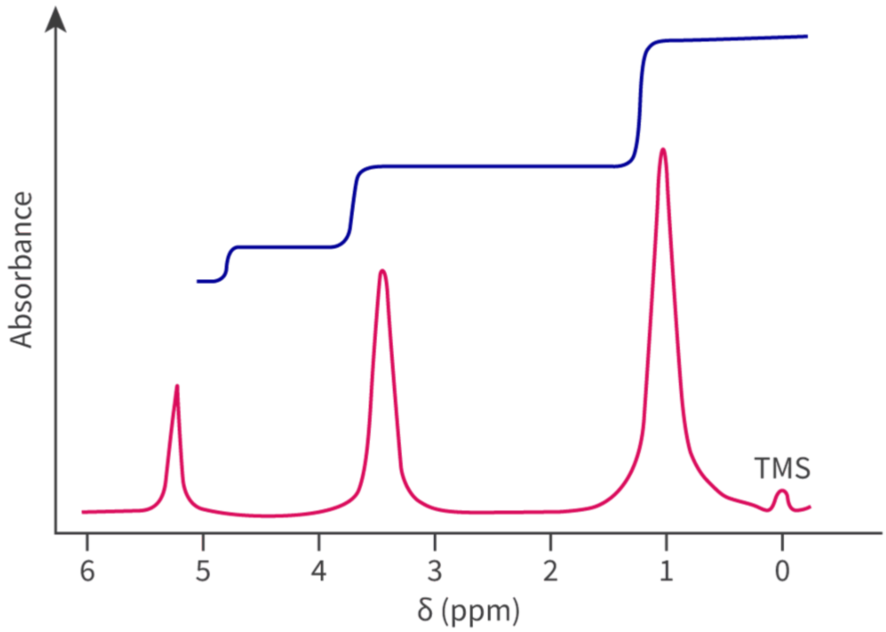


Figure 13. The ^{1}H NMR spectrum of a compound.

More information for figure 13

The image is a 1H NMR spectrum graph. The x-axis represents the chemical shift in parts per million (ppm), ranging from 0 to 6 ppm, labeled with specific markers. The y-axis indicates absorbance, though specific units are not shown. The spectrum displays a series of peaks, with significant peaks occurring around 1 ppm, 3 ppm, and 5 ppm. There is a baseline offset shown with a blue line, indicating integrations or sums of the peak areas. The peaks

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indicate the presence and environment of hydrogen atoms within the molecular structure of the sample being analyzed, revealing structural information about the compound. A small peak labeled "TMS" appears near 0 ppm, serving as a reference point.

[Generated by AI]

1. There are three separate signals indicating three different chemical environments containing protons.
2. The integration trace at the top can be measured with a ruler and the heights compared relative to each other. The ratio of protons in each environment is 1:2:3.
3. Using section 21 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/%C2%B9h-nmr-data-id-45123/) of the DB chemistry data booklet, the range for the chemical shift and the number of protons in each environment can be used together to identify the protons likely to be responsible for each signal. The signal at chemical shift 1.0 contains three protons, indicating the presence of a methyl group (—CH3), which gives a signal typically in the 0.9-1.0 ppm range. The signal at chemical shift 5.2 contains one proton, likely from the hydroxyl group (—OH) in an alcohol, which typically gives a signal in the 1.0—6.0 ppm range.

The use of TMS as the reference standard

The chemical shift is measured relative to tetramethylsilane, which is usually abbreviated to TMS. The structure of TMS is shown in **Figure 14**.

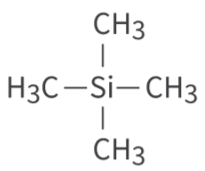

Figure 14. The full structural formula of tetramethylsilane (TMS).

More information for figure 14

TMS is assigned a chemical shift of 0 ppm and is used as the reference standard for the following reasons:

- It is unreactive and non-toxic.
- It is volatile, meaning that it has a low boiling point. For this reason, it can easily be removed from the sample being analysed.

- All the protons in TMS are equivalent (in the same chemical environment), therefore, it produces a strong single signal (or peak) upfield, away from other signals.

It should be noted that the peak for TMS is not counted to determine the number of different chemical environments that contain protons because it is not part of the molecule under investigation.

Splitting patterns

So far, we have looked at low resolution ^{1}H NMR spectra. A high resolution ^{1}H NMR spectrum provides additional information about an unknown organic compound. When observing a high resolution ^{1}H NMR spectrum we see that the peaks produced by the different chemical environments are split into clusters of peaks (multiplets), which are known as splitting patterns. This is a result of the interactions between the magnetic fields of adjacent non-equivalent protons, known as spin–spin coupling. The splitting patterns covered in the DP chemistry course are shown in **Figure 15**.

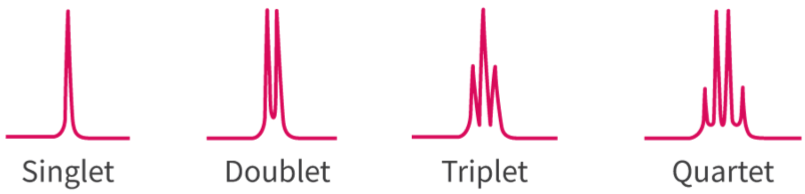

Figure 15. The splitting patterns produced as a result of spin—spin coupling.

More information for figure 15

The image displays four NMR splitting patterns, each labeled below with its corresponding name. From left to right, the patterns are: Singlet, Doublet, Triplet, and Quartet.\n\n1. **Singlet**: A single tall peak indicates the absence of adjacent non-equivalent protons.\n2. **Doublet**: Two peaks of similar height indicate one adjacent non-equivalent proton, demonstrating the spin-spin coupling effect.\n3. **Triplet**: Three peaks of similar height show the presence of two adjacent non-equivalent protons.\n4. **Quartet**: Four peaks of similar height appear due to three adjacent non-equivalent protons.

[Generated by AI]

To deduce the splitting pattern produced, count the number of non-equivalent protons on the adjacent carbon atom and add one to that number, which is known as the n+1 rule, (where *n* is the number of non-equivalent protons). The *n*+1 rule is used together with Pascal's triangle to predict the splitting patterns on a ^{1}H NMR spectrum. **Figure 16** shows a summary of this, together with the relative intensities of the protons.

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Number of protons bonded to adjacent carbon atom	Relative intensity							Splitting pattern
$n = 0$				1				singlet
$n = 1$			1		1			doublet
$n = 2$		1		2		1		triplet
$n = 3$	1		3		3		1	quartet

Figure 16. A summary of the number of adjacent protons, the relative intensities and the splitting patterns covered in this section.

More information for figure 16

The table displays the relationship between the number of protons bonded to an adjacent carbon atom, the relative intensities, and the splitting patterns in a proton NMR spectrum. The categories are as follows:

1. For n = 0 protons, the relative intensity is 1, and the splitting pattern is a singlet.
2. For n = 1 proton, the relative intensities are 1 for each proton, resulting in a doublet splitting pattern.
3. For n = 2 protons, the relative intensities are 1, 2, and 1, creating a triplet splitting pattern.
4. For n = 3 protons, the relative intensities are 1, 3, 3, and 1, leading to a quartet splitting pattern.

Each row provides the splitting pattern details depending on the number of adjacent protons (n) involved.

[Generated by AI]

The high resolution ^{1}H NMR of ethanal is shown in **Figure 17**. There are two clusters of peaks: one cluster for the $-CH_3$ group (a doublet) and one cluster for the $-CHO$ group (a quartet).

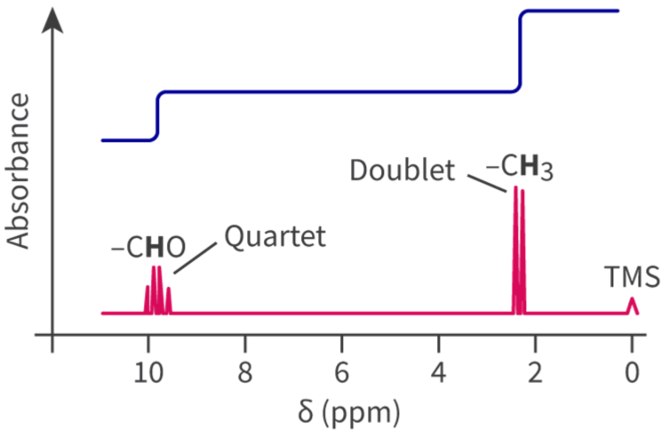


Figure 17. The high resolution ^{1}H NMR spectrum of ethanal shows two clusters of peaks.

More information for figure 17

The image is a graph depicting the high-resolution 1H NMR spectrum of ethanal. The X-axis represents the chemical shift (δ) in parts per million (ppm), ranging from 0 to 12 ppm, while the Y-axis is labeled 'Absorbance'. There are two key clusters of peaks. The first, closer to 10 ppm, is labeled '-CHO Quartet'. The second, closer to 2 ppm, is labeled 'Doublet -CH3'. The graph also displays a small peak at 0 ppm labeled 'TMS', which is typically used as a reference signal in NMR spectroscopy. The spectrum indicates how magnetic fields affect the protons in different chemical environments, such as the influence of the CH3 group on the adjacent CHO group and vice versa.

[Generated by AI]

The magnetic field experienced by the protons in the $-CH_3$ group is affected by the one proton on the adjacent carbon atom of the –CHO group. Using the n+1 rule, the splitting pattern is a doublet. The magnetic field experienced by the proton in the –CHO group is affected by the three protons in the $-CH_3$ group. Again, we use the n+1 rule to determine the splitting pattern, which is a quartet (**Figure 18**).

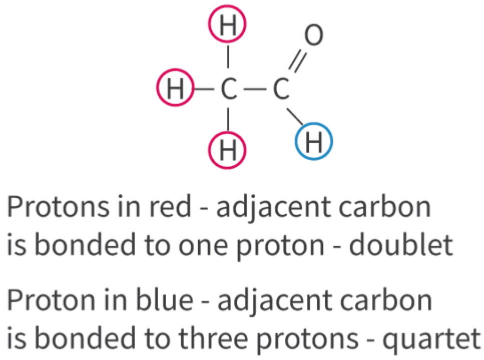


Figure 18. The splitting patterns in ethanal are a doublet and a quartet.

More information for figure 18

The image is a chemical diagram depicting the molecular structure of ethanal. It illustrates the arrangement of atoms and the splitting patterns observed in nuclear magnetic resonance (NMR) spectroscopy. In the diagram, hydrogen atoms (protons), depicted as "H," are bonded to a carbon atom, represented by "C." One carbon is double-bonded to an oxygen atom, indicated by "O," completing the carbonyl group.

The diagram highlights two splitting patterns in protons, shown with color coding. Protons in red are bonded to the second carbon, while one adjacent proton on the other carbon forms a doublet splitting pattern. The text next to the diagram explains: "Protons in red - adjacent carbon is bonded to one proton - doublet."

Protons in blue, bonded to the carbon attached to the three hydrogens, form a quartet splitting pattern due to the presence of three neighboring protons. The text reads: "Proton in blue - adjacent carbon is bonded to three protons - quartet."

This explanation correlates with the "n+1" rule in NMR where the number of peaks in a splitting pattern is determined by the number of neighboring protons plus one.

[Generated by AI]

The last example we will look at in terms of splitting patterns is the compound ethanol, CH_3CH_2OH. How many signals would you expect to observe on its ^{1}H NMR spectrum and what would the splitting patterns be?

The high resolution ^{1}H NMR spectrum is shown in **Figure 19**. Is it what you expected?

The spectrum shows a quartet (which is caused by the three protons in the CH_3 group), a triplet (caused by the two protons in the CH_2 group) and a singlet. The reason for the singlet is that the proton of the OH group does not interact with adjacent protons unless the alcohol is very pure without any traces of water. So, the splitting of the CH_2 group is only affected by the protons on the adjacent CH_3 group (producing a quartet) and not by the H of the OH group. In addition, the CH_2 group has no effect on the splitting of the proton of the OH group, hence the singlet. The ratio of the protons in each different chemical environment can be determined (1:2:3), which provides useful information about the compound.

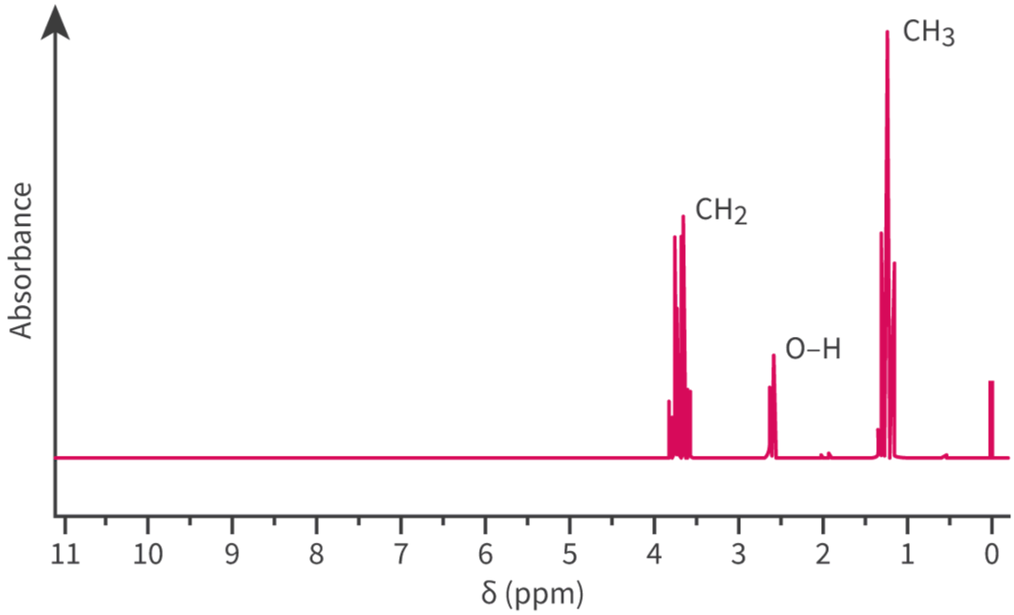


Figure 19. The splitting patterns in ethanol are not what you might expect.

More information for figure 19

The image is a graph depicting the NMR (Nuclear Magnetic Resonance) spectrum of ethanol. The X-axis is labeled with chemical shift in δ (ppm) ranging from 11 to 0. The Y-axis represents absorbance.

The spectrum shows three main peaks: 1. A quartet corresponding to the CH3 group, visible around δ = 1.2 ppm. 2. A triplet corresponding to the CH2 group, seen around δ = 3.7 ppm. 3. A singlet that represents the O-H proton, appearing near δ = 5.5 ppm.

The triplet and quartet arise due to spin-spin splitting from interactions between neighboring protons while the singlet is due to the absence of such interactions, often related to proton exchange in the O-H group especially when impurities like water are absent.

[Generated by AI]

Worked example 4

Deduce the splitting patterns observed on the ^{1}H NMR spectrum of methyl ethanoate.

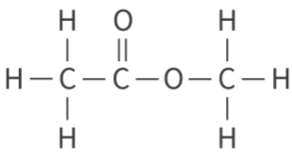


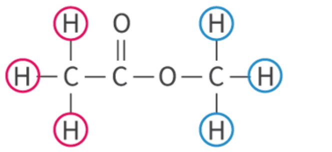
Figure 20. Methyl ethanoate.

More information for figure 20

The image is a chemical structure diagram of methyl ethanoate. It shows the molecular structure represented with bonded lines between atoms. At the center is an ethanoate group, with a carbon atom double-bonded to an oxygen atom and single-bonded to another oxygen atom which is also bonded to a methyl group (CH3). The central carbon is also bonded to a methyl group. The diagram shows the orientation and connectivity of each atom, typical of a structural formula in chemistry, with lines representing chemical bonds and the arrangement of the atoms indicating the molecular geometry.

[Generated by AI]

The protons in the two CH_3 groups are non-equivalent so methyl ethanoate has two different chemical environments that contain protons. Applying the *n*+1 rule, the protons in the CH_3 group on the left would produce a singlet as would the protons in the CH_3 group on the right. So the splitting patterns are two singlets.



In this activity, you will make predictions about the ^{1}H NMR spectra for a variety of organic molecules.

Activity

- **IB learner profile attribute:** Knowledgeable

- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. For each of the following compounds, deduce the number of different chemical environments that contain protons.

   (a) $CH_3CH_2CH_2CH_3$
   (b) $CH_3CH_2CH_2OH$
   (c) $CH_3COOCH_2CH_2CH_3$
   (d) $CH_3CH_2NH_2$
   (e) $CH_3COCH_2CH_2CH_2CH_3$
   (f) $CH_3CH_2CH_2Cl$
   (g) C_6H_6

2. For each of the compounds in question 1, deduce the ratio of protons in each different chemical environment.
3. For each compound, deduce the splitting patterns observed.

   (a)

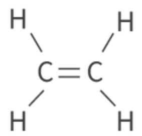


Figure 21. Deduce the splitting patterns observed.

More information for figure 21

The image shows the structural formula of an ethylene (C2H4) molecule. At the center, two carbon atoms are doubly bonded together, represented by two parallel lines (C=C). Each carbon atom is bonded to two hydrogen atoms, represented by lines connecting to the hydrogen symbols (H). The hydrogen atoms are symmetrically positioned around the carbon atoms, illustrating the planar geometry typical for ethylene. This diagram highlights the sp2 hybridization and planar geometry of the ethylene molecule.

[Generated by AI]

(b)

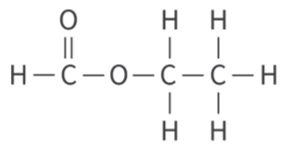


Figure 22. Deduce the splitting patterns observed.

More information for figure 22

The image shows a chemical structure diagram representing a molecular compound. At the top, an oxygen (O) atom is double-bonded to a carbon (C) atom, forming part of a carbonyl group. This carbon is also single-bonded to another carbon atom through an oxygen bridge, indicating an ester linkage. The second carbon atom is connected to two hydrogen (H) atoms. This part is followed by a similar structure on the right, where the carbon is bonded to more hydrogen atoms. The overall structure appears linear and illustrates a simple organic compound with hydrogen atoms fulfilling remaining valencies around the carbon atoms.

[Generated by AI]

1. (a) $CH_3CH_2CH_2CH_3$ Two different chemical environments that contain protons
   (b) $CH_3CH_2CH_2OH$ Four different chemical environments that contain protons
   (c) $CH_3COOCH_2CH_2CH_3$ Four different chemical environments that contain protons
   (d) $CH_3CH_2NH_2$ Three different chemical environments that contain protons
   (e) $CH_3COCH_2CH_2CH_2CH_3$ Five different chemical environments that contain protons
   (f) $CH_3CH_2CH_2Cl$ Three different chemical environments that contain protons
   (g) C_6H_6 One chemical environment that contain protons
2. (a) $CH_3CH_2CH_2CH_3$ Ratio of protons 4:6
   (b) $CH_3CH_2CH_2OH$ Ratio of protons 1:2:2:3(c) $CH_3COOCH_2CH_2CH_3$ Ratio of protons 2:2:3:3
   (d) $CH_3CH_2NH_2$ Ratio of protons 2:2:3
   (e) $CH_3COCH_2CH_2CH_2CH_3$ Ratio of protons 2:2:2:3:3
   (f) $CH_3CH_2CH_2Cl$ Ratio of protons 2:2:3
   (g) C_6H_6 Ratio of protons N/A (only one chemical environment)
3. (a) Singlet (there are two sets of equivalent protons so no splitting occurs).
   (b) Singlet, triplet, quartet

Kog